

3.3 Mission Task Descriptions

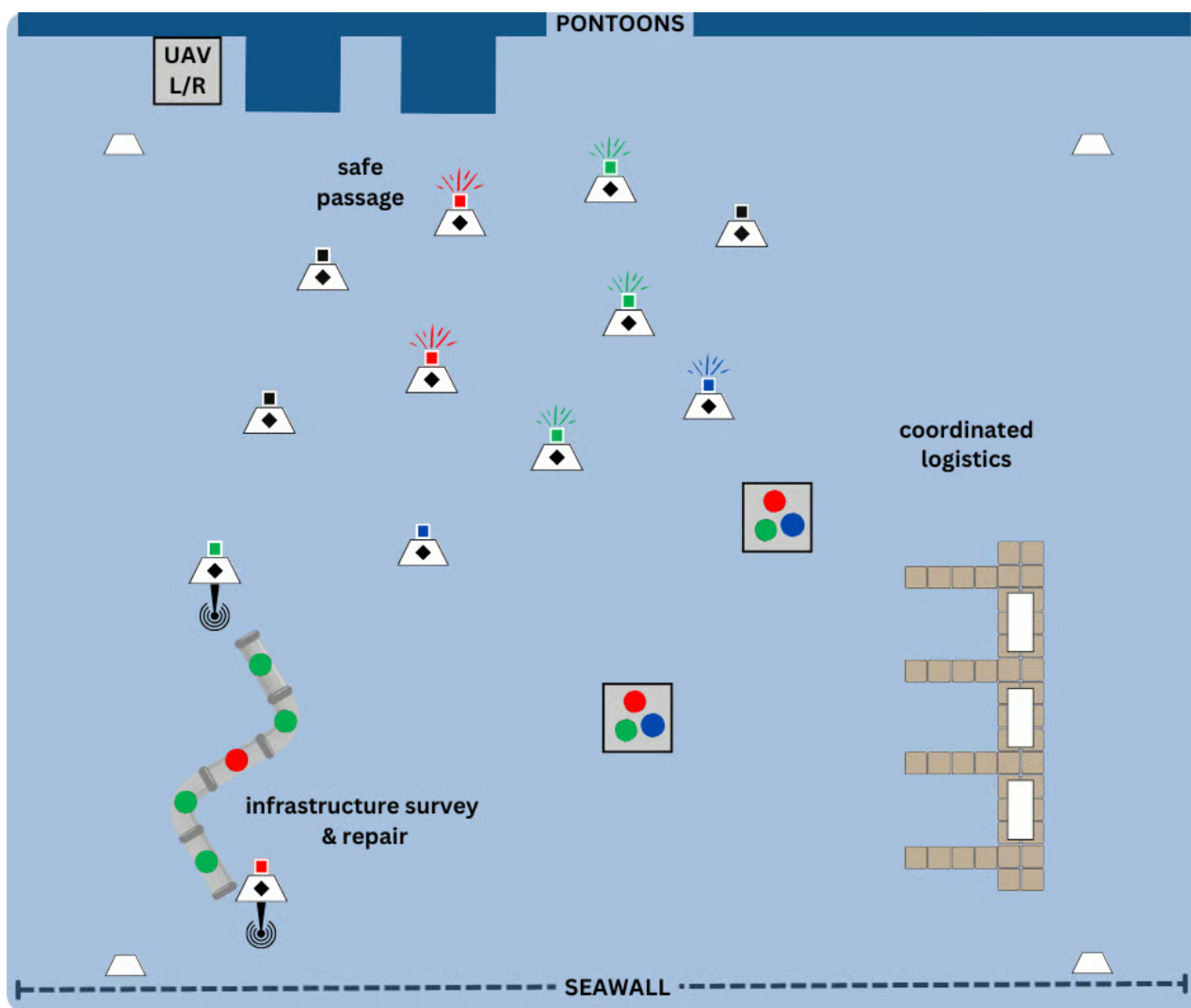
This section provides details of the RobotX 2026 Autonomy Challenge tasks and the proficiency requirements for each round of competition. Teams are encouraged to develop a strategy to approach these tasks that best suits their SoS.

All uncrewed systems must be switched to autonomous mode to initiate start of any collaborative autonomous run in the courses.

Capability Levels

Each mission task has capabilities divided by three levels:

- **Core Capabilities:** Fundamental competencies required for safe and effective baseline autonomous operation.
- **Advanced Capabilities:** Competencies that reflect growth in autonomy sophistication and real-world relevance.
- **Disruptive Capabilities:** Transformative competencies pushing the boundary of autonomous systems - enabling novel or mission-critical applications.



Preliminary Course Layout

3.3.1 Cross-cutting Tasks

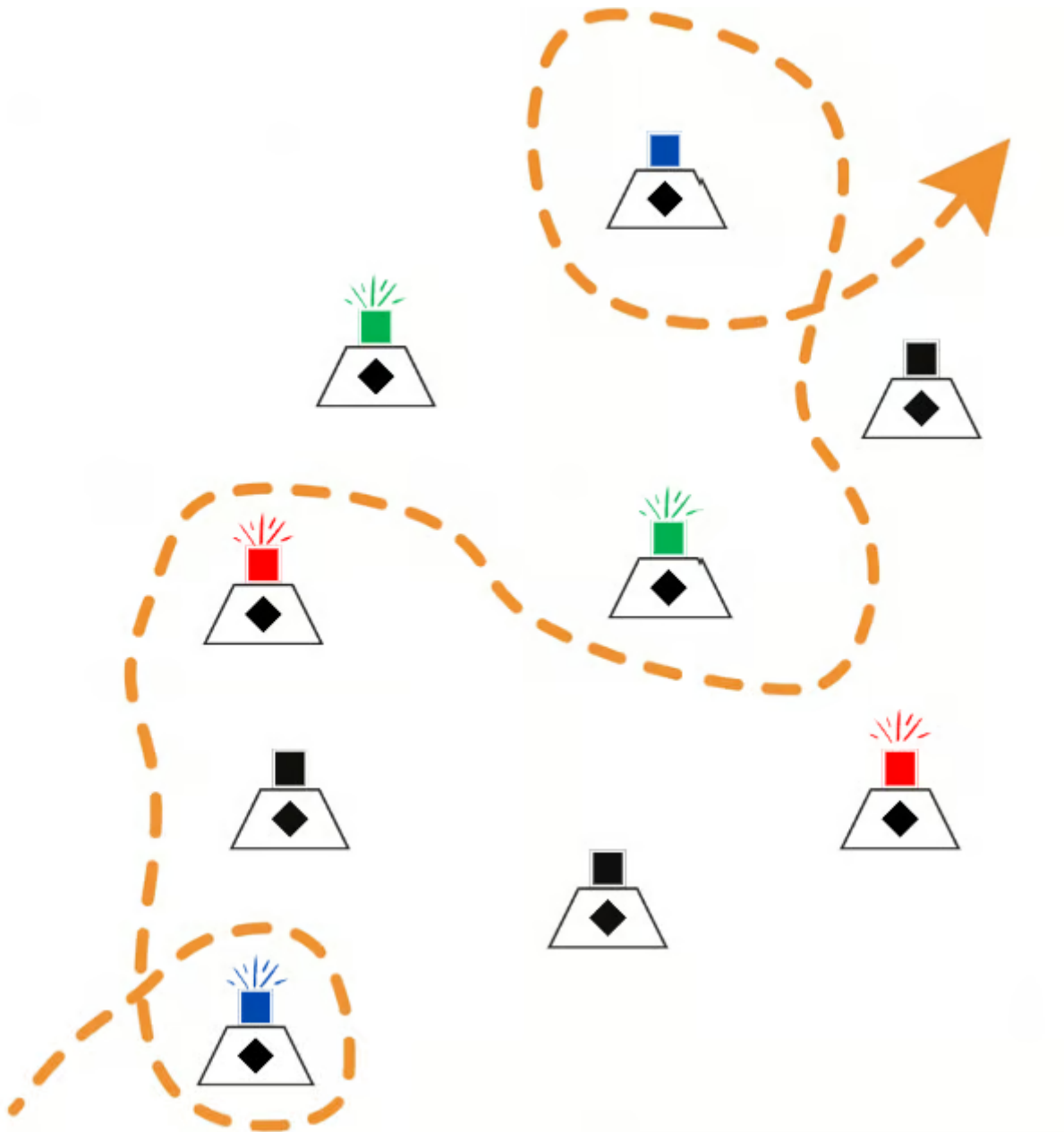
All uncrewed systems are required to autonomously perform basic functions as part of mission execution and safe operation. These functions cut across all aspects of autonomous mission execution.

- **RoboCommand:** The System of Systems (SoS) must provide updates and reporting for each vehicle operating during any run in the autonomy challenge courses via the RoboCommand system. This is to allow competition organizers, technical teams, and judges to have good situational awareness of the SoS and to evaluate performance against the challenges.
- **Obstacle Avoidance:** The ability to avoid obstacles is a critical capability for safe operation of uncrewed systems. Every course element is an object to be avoided or approached safely in some way. In addition, obstacle buoys may be placed throughout the operating areas in an effort to provide a more representative real-world challenge.
- **Inter-Vehicle Communication:** Inter-vehicle communication is a central component to multi-vehicle collaborative execution of the tasks as required by Advanced and Disruptive Tiers. Teams are encouraged to provide clear and easily understandable demonstration of this capability for judges to understand.
- **Dynamic Adaptability:** In real-world scenarios, uncrewed systems must be able to adapt to changes in the operating environment or other mission priority changes. [Mission Task 4](#) represents these conditions through use of the RoboCommand system to transmit dynamic changes to the mission priorities or operating environment.

3.3.2 Mission Task 1: Safe Passage

Minimum Systems Required: USV (or surfaced UUV), UAV

The maritime systems must safely enter the recovery area in order to begin survey and relief efforts. There are hazards to surface and underwater vessels, represented by buoys topped with light beacons in the transit area.



Example Task 1 Layout and USV Path

This task consists of a group of ten (10) buoys with LED light beacons that will be in one of five states: OFF, flashing **RED**, flashing **GREEN**, flashing **BLUE**, and steady **BLUE**. The flashing lights will repeatedly cycle on for 1 second and off for 1 second. These represent objects on the water that must be approached and avoided as described below.

The buoys will have two light beacons, one visible from the side (for surface systems), and one visible only from above (for the UAV). Only one of these beacons will be active at any time.

Each buoy will have lights visible from the side (for surface systems) or only from above (for the UAV). Depending on the tier to be attempted, one or both of these beacons will be activated to show a safe path through the area.

When a System of Systems (SoS) reports that it is attempting an autonomous passage, the task will activate light beacons in a random configuration based on the tier to be attempted. The task will remain active for as long as a system continues to report attempts at the task.

A surface system (USV or UUV) must autonomously transit on the water surface from the correct ENTRY point to the correct EXIT point while avoiding obstacles and passing the **RED** and **GREEN** buoys on the appropriate side of the surface system.

- A **FLASHING RED** light must be passed on the surface system's starboard side.
- A **FLASHING GREEN** light must be passed on the surface system's port side.
- A **FLASHING BLUE** light indicates the safe ENTRY point for the surface system.
- A **SOLID BLUE** light indicates the safe EXIT point for the surface system.
- A **BLACK** indicator refers to one in which the light beacon is not active; this is an obstacle to transit and may be passed to any side.

In the Advanced and Disruptive Tiers the UAV must complete an overfly of the start area and report via the RoboCommand system to determine a safe passage before the USV can transit into the field to attempt safe passage. Light beacons visible only from above will be placed atop buoys to indicate a safe route. The USV will then be required to transit the course identified by the UAV. Safe route changes in a random pattern for each run.

Core Tier

The System of Systems (SoS) must identify and report the correct start and end points for the transit, as well as the **RED**, **GREEN**, and **BLACK** (OFF) buoys as detected and classified. The USV or UUV must approach and fully circle around the ENTRY buoy with the **FLASHING BLUE** light in a **clockwise direction** before beginning transit of the buoy field.

The USV or UUV must transit through the area on the surface after circling the ENTRY buoy and exiting near the correct EXIT marker while passing the **RED/GREEN** buoys on the correct side. The **RED/GREEN** buoys will have lights visible from the side at the Core Tier.

Once the USV or UUV has completed transit, the craft must fully circle around the EXIT buoy with the **SOLID BLUE** light in a **counterclockwise direction** to complete this task.

Capabilities Demonstrated:

- UAV (optional): Autonomous search, detect, localize, and ID light beacons, Report to RoboCommand, Communicate with other systems.
- USV/UUV: Detect, localize, and ID light beacons, Autonomous path planning, Autonomous path following, Avoid obstacles, Report to RoboCommand.

Advanced Tier

At the Advanced Tier, the safe passage is only detectable by the UAV, which must relay the ENTRY and EXIT points, as well as the location of **RED** and **GREEN** markers, to the surface system and to RoboCommand. The USV or UUV must use the data from the UAV to safely transit the field. The path will be randomized for each run, but will remain constant for the duration of a single attempt. Thus, the UAV is not required to remain on station tracking passage of the surface system.

Capabilities Demonstrated:

- UAV: Autonomous search, detect, localize, and ID light beacons, Report to RoboCommand, Communicate with other systems.
- USV/UUV: Detect buoys, Path planning/execution based on UAV shared data, Avoid obstacles, Receive tasking from UAV.

Disruptive Tier

The disruptive tier builds on the Advanced Tier by adding the possibility that the safe passage may change during the surface system transit through the field of buoys. This represents detection by the UAV of a previously not seen hazard to navigation.

Changes in the hazards will be visible only to the UAV, which means that the UAV will need to remain on station to provide updates to the surface system in real time, demonstrating the ability of the SoS to communicate and adapt to the changing environment.

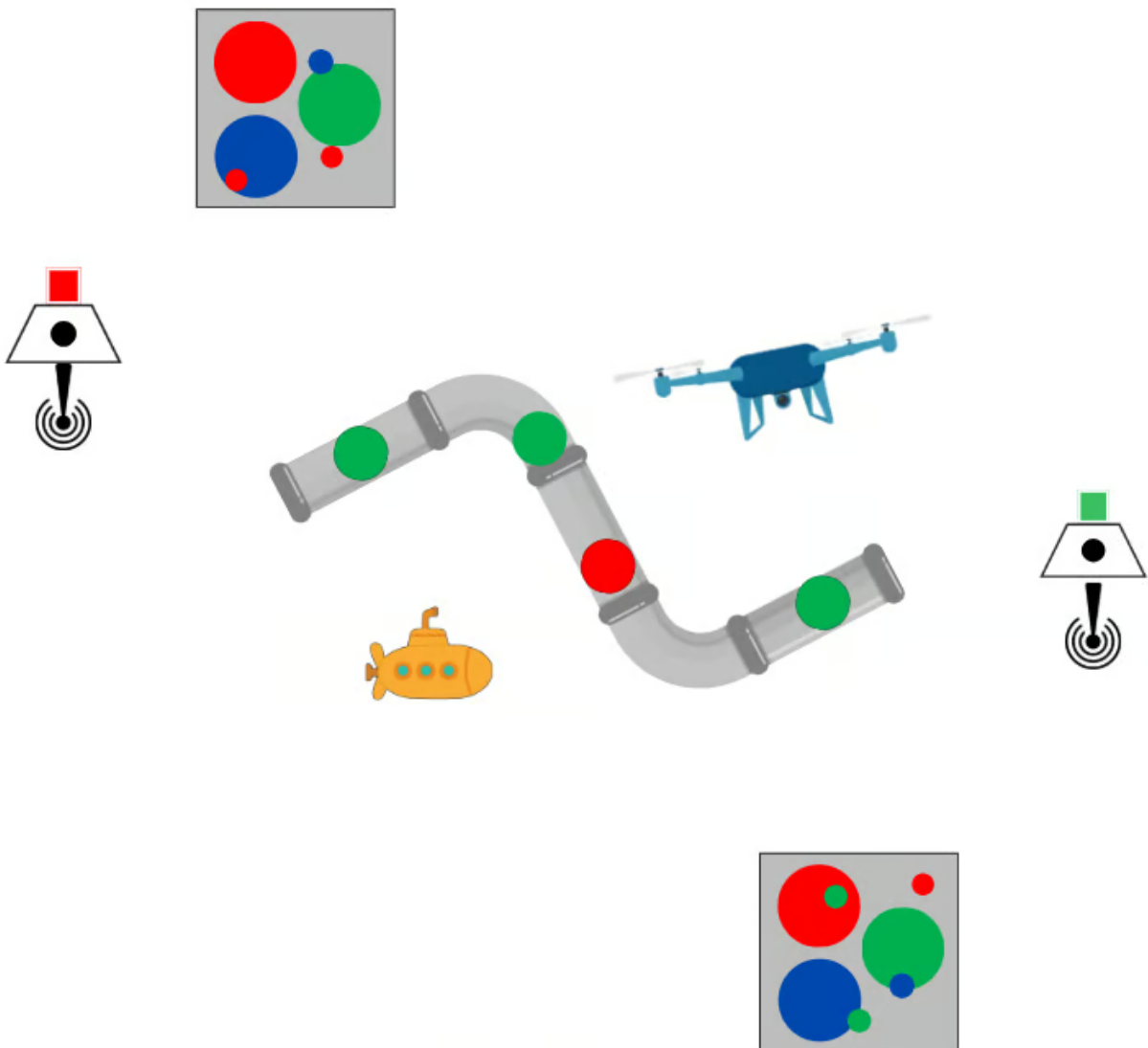
Capabilities Demonstrated:

- UAV: Autonomous search, detect, localize, and ID light beacons, Persistent surveillance, Detect changes, Report to RoboCommand (detections and changes), Communicate with other systems.
- USV/UUV: Detect buoys, Path planning/execution based on UAV shared data, Avoid obstacles, Receive tasking and re-tasking from UAV.

3.3.3 Mission Task 2: Infrastructure Survey and Repair

Minimum Systems Required: UUV, UAV

An underwater pipeline is indicating signs of damage in the harbor. It must be surveyed for damage and repaired. The area to be surveyed will be indicated on the surface by a pair of buoys.

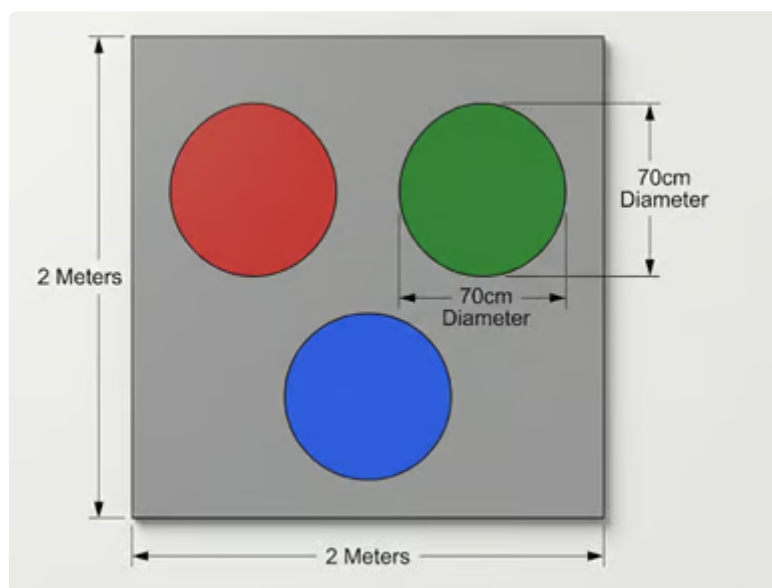


Mission Task 2: Infrastructure Survey and Repair

The task will consist of surface and underwater course elements. A pipeline will be suspended underwater approximately 1-2 meters above the seafloor. The pipeline will be equipped with LED ring lights that change between **GREEN** and **RED** to indicate functional and damaged areas, respectively. At the center of each light ring will be a magnetic switch that turns the **RED** light to **GREEN**.

Two buoys will be placed near the ends of the pipeline. Each buoy will have a color indicator on top and an underwater acoustic pinger suspended below. The color indicators will change between **RED** and **GREEN**. Only the acoustic pinger suspended below **GREEN** buoy will be actively pinging.

There will also be two floating platforms within the course boundary that contain **RED**, **GREEN**, and **BLUE** tins. Each of the floating platforms will be approximately 2 meters square, covered with a grey background with **RED**, **GREEN**, and **BLUE** circles on it. The circles will be 60-70cm in diameter. These platforms are also used in Task 3.



Example UAV Delivery Platform

The colors for the background and colored circles will be as near as we are able to produce to those listed in the table below.

Platform Element	Hex Code	Pantone Color
Background Color (Benjamin Moore Sidewalk Grey ↗)	#CCCED0	179-3 C
Red Circle	#E8282B	1788 C
Green Circle	#089404	355 C
Blue Circle	#0072CE	285 C

Tins colored **RED**, **GREEN**, and **BLUE** will be used to represent the resources to be delivered by UAVs. The tins will be weighted to approximately 40-50 grams and coated using a spray Plasti-Dip to help maintain their watertightness and durability. The components in the table below will be used to make the resource tins.



Uncoated Aluminum Tin Dimensions

Task Element	Description	Source
Metal Tins (~40-50 grams)	Aluminum Round Cans	https://a.co/d/03wsKktP ↗
Red Plastic Coating	Plasti-Dip Spray - Red	https://a.co/d/0iZDJoFA ↗
Green Plastic Coating	Plasti-Dip Spray - Blaze Green	https://a.co/d/01smaPAH ↗
Blue Plastic Coating	Plasti-Dip Spray - Blaze Blue	https://a.co/d/04vVGhNe ↗

When the challenge activates, one of the color indicators on the buoys will turn **GREEN** and the acoustic pinger on the same buoy will begin pinging. The UUV should begin its survey to locate the pipeline at this active buoy.

Simultaneously, the underwater pipeline will activate, illuminating one of the lights **RED** while the remaining lights turn **GREEN**. The **RED** light indicates the section in need of repair, and it will also indicate the area where the UUV must hold a magnetic probe to represent performing a repair of the damaged section.

Core Tier

The UUV must transit to the survey area start point to begin execution of the challenge. This start point is identified by a **GREEN** color indicator on the buoy. Once the UUV reaches the search area, it must find the underwater pipeline and follow it, noting whether each segment is intact (**GREEN**) or damaged (**RED**).

The minimum response at the Core Tier is a report of the pipeline status to the RoboCommand system. The order of reporting begins with the part of the pipeline closest to the active buoy. The UUV may surface to complete this reporting.

For additional points the UUV may “repair” the damaged segment by holding a magnetic probe in the area indicated near the light. Once the marker is held in place long enough, the pipeline will notify RoboCommand and change the light to **GREEN**.

Capabilities Demonstrated:

- UAV: Identify and localize the **GREEN** indicator, communicate start point to UUV for task execution.
- USV: Identify and localize the **GREEN** indicator or localize the active acoustic pinger, communicate start point to UUV for task execution.
- UUV: Identify **GREEN** indicator or home to acoustic pinger, Locate underwater pipeline, follow pipeline, report light pattern, precisely position probe to "repair" pipeline segment, report to RoboCommand.

Advanced Tier

The Advanced Tier builds on the Core Tier by requiring the UUV to request a supply delivery by the UAV after completing the repair of the pipeline segment. Once the UUV is on site, it conducts the Core Tier mission by surveying AND repairing the pipeline segment.

After the UUV has successfully completed its repair of the pipeline segment, the **RED** light will change to **GREEN** for five (5) seconds, it will then flash **RED**, **GREEN**, or **BLUE** to indicate where a resource needs to be delivered on one of the floating platforms. The flashing light will be on for one (1) second, then off for one (1) second, repeating this pattern for 30 seconds before turning off.

The UUV must transmit this resource request to the UAV and report successful task completion to RoboCommand.

The UAV must acknowledge the re-tasking by reporting a tasking change to RoboCommand and then deliver ANY colored tin to the correct circle on the floating platform. At this tier, the UAV may be pre-loaded with a tin, but must already be in autonomous mode when the UUV sends its message.

In order to help judges understand what is happening, the UUV and UAV must report their status and progress in accordance with the RoboCommand messages described in [Section 3.4 Communications Protocol](#).

Capabilities Demonstrated:

- UAV: Receive tasking from UUV, Report to RoboCommand, Deliver payload, Identify and localize the **GREEN** indicator, communicate start point to UUV for task execution.
- USV: Identify and localize the **GREEN** indicator or localize the active acoustic pinger, communicate start point to UUV for task execution.
- UUV: Identify **GREEN** indicator or home to acoustic pinger, Locate underwater pipeline, follow pipeline, report light pattern, precisely position probe to "repair" pipeline segment, report to RoboCommand, communicate with UAV.

Disruptive Tier

The Disruptive Tier builds on the Advanced Tier by adding an extra flash of the lights on the pipeline, telling the UUV both what resource is needed and to which area it should be delivered.

After the UUV has successfully completed its repair of the pipeline segment, the **RED** light will change to **GREEN** for five (5) seconds, then flash two colors, **RED**, **GREEN**, or **BLUE** to indicate which resource is needed and to which color it should be delivered. This randomly generated a two-color sequence will show the 1st color for one (1) second, then the off for one (1) second, then the 2nd color for one (1) second, the off (**BLACK**) for 2 seconds. The same pattern is repeated continuously for 30 seconds.

The first color indicates which color tin is needed and the second color indicates to which colored circle the tin should be delivered.

Once the UUV completes its repair and receives the resource request, it may surface to report to RoboCommand and transmit the request to the UAV for completion. The UAV must then locate and retrieve the correct colored tin and successfully deliver it to the correct resource circle.

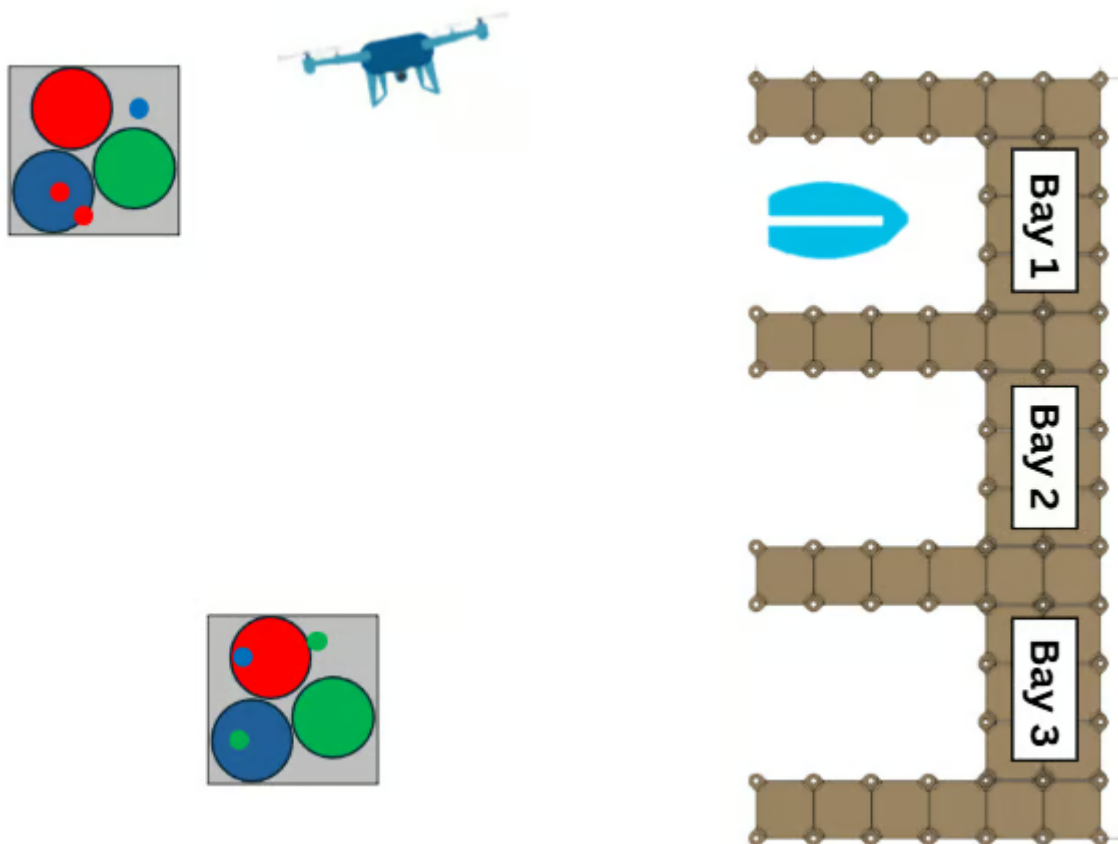
Capabilities Demonstrated:

- UAV: Receive tasking from UUV, Report to RoboCommand, Retrieve and deliver payload, Identify and localize the **GREEN** indicator, communicate start point to UUV for task execution.
- USV: Identify and localize the **GREEN** indicator or localize the active acoustic pinger, communicate start point to UUV for task execution.
- UUV: Identify **GREEN** indicator or home to acoustic pinger, Locate underwater pipeline, follow pipeline, report light pattern, precisely position probe to "repair" pipeline segment, report to RoboCommand, communicate with UAV.

3.3.4 Mission Task 3: Coordinated Logistics

Minimum Systems Required: USV (or surfaced UUV), UAV

This task represents multiple scenarios that may exist after a natural disaster: (a) the USV must find the safe docking bay, (b) complete a "firefighting task", and (c) request resources such as food, medical, or other supplies be delivered. The uncrewed systems can work as a team to quickly complete these challenges.



Example of the Task3 Docking Bays and Delivery Platforms

The task will consist of floating docks assembled to provide three docking bays, with bay numbers referenced left-to-right when facing the bays (as shown). Within each bay there will be a small, white structure that represents a building. Each structure will have two 25cm square target holes with black borders, and at the base of the face there will be a color indicator showing either **RED** or **GREEN** to indicate which bay is available for docking.

This task will also use the two floating platforms and resource tins described in [Task 2](#). The UAV will be required to specify for which task it is executing the resource delivery in order to be eligible to score points for the Advanced and Disruptive tiers.

The surface craft must locate and dock in the available docking bay, indicated by a **GREEN** color indicator, then report a successful docking to the RoboCommand system. The bays with the **RED** color indicator are not safe bays in which to dock. When the USV reports successful docking, the rest of the challenge will activate as described in the sections below. The USV will be required to communicate with the UAV to enable the air delivery portion of the task (Advanced and Disruptive Tiers).

In the Advanced and Disruptive Tiers one of three (3) resources, represented by **RED**, **GREEN**, and **BLUE** tins, must be delivered from one location to another. The UAV launch area and one or two floating platforms on the water will be the origin points for resources (depending on challenge tier), and the delivery area will be at one of the floating platforms.

Core Tier

The USV must locate the floating docking platform, fully dock in the correct bay, identified by the **GREEN** indicator, then report to the RoboCommand system that it has successfully docked.

When the USV reports to RoboCommand that it has successfully docked, a **RED** light inside one of the windows within that bay will activate, indicating the active target area. The USV must then spray water into the correct window until the light turns **GREEN**. The **GREEN** light will stay on for five (5) seconds before turning off. When this is complete, the USV must report successful completion of the Core Tier challenge.

Capabilities Demonstrated:

- UAV: No tasking.
- USV: Detect safe docking bay, Detect correct window, Deliver water to correct window, Detect successful water delivery, Report to RoboCommand.
- UUV: No tasking.

Advanced Tier

The USV must complete the Core Tier by successfully docking and "putting out the fire" as described in the Core Tier. The light will turn **GREEN** for five (5) seconds as in the Core Tier, then begin flashing (1 second on, 1 second off) the color of the circle to which the UAV should deliver its "relief supply" payload (**RED**, **GREEN**, or **BLUE**). The flashing will continue for 60 seconds, after which the light will turn off.

The USV must relay the delivery destination to the UAV for fulfillment and report this information to RoboCommand for maximum points.

The UAV earns points by delivering ANY tin to the correct colored circle on the delivery platforms. The UAV may be pre-loaded with a tin at this tier, but will earn more points for retrieving and delivering a tin autonomously.

The USV does NOT need to remain in the dock while the UAV completes its part of the task.

Capabilities Demonstrated:

- UAV: Respond to tasking from USV, Locate and retrieve any resource, Deliver resource to correct target area.
- USV: Detect safe docking bay, Report to RoboCommand, Interpret resource request, Communicate with UAV.
- UUV: No Tasking.

Disruptive Tier

As in the Core and Advanced Tiers, the USV must first locate and dock in the safe bay and report readiness to receive the next task from RoboCommand. The USV must put out the fire, causing the target window light to change from **RED** to **GREEN**.

The light will remain **GREEN** for five (5) seconds, turn off for one (1) second before it begins flashing the color of the needed resource AND the color of the receiving bay where it needs to be delivered (**RED**, **GREEN**, or **BLUE**). This randomly generated two-color sequence will show the 1st color for one (1) second, then off for one (1) second, then the 2nd color for one (1) second, then off (**BLACK**) for 2 seconds. The same pattern is repeated continuously for 60 seconds.

The USV must report the detected pattern to RoboCommand and relay the resource request to the UAV for maximum points.

The UAV must either launch from the start point or divert from its current tasking, locate the platform with the correct resource (colored tin), retrieve the resource, and deliver it to the appropriate colored circle on the other platform.

Capabilities Demonstrated:

- UAV: Respond to tasking from USV, Locate and retrieve **CORRECT** resource, Deliver resource to correct target area.
- USV: Detect safe docking bay, Report to RoboCommand, Interpret resource request, Communicate with UAV.
- UUV: No Tasking.

3.3.5 Mission Task 4: Dynamic Incident Response

Minimum Systems Required: USV, surfaced UUV, UAV

In real-world scenarios, uncrewed systems must be able to adapt to changes in the operating environment or other mission priority changes. These conditions will be represented through use of the RoboCommand system to broadcast a new task

Core Tier

The RoboCommand system will transmit a request for assistance at a specific GPS position for a specific domain (air or surface). The uncrewed system operating in that domain must acknowledge receipt and intent to support via the appropriate RoboCommand message.

The specified system must immediately acknowledge re-tasking, break off of its current task, transit to the requested area, loiter, and report readiness for further tasking. The RoboCommand system will respond to the readiness message indicating receipt of the message from the uncrewed system and clearance to return to normal operations.

In the core tier, the system may resume normal operation after receiving confirmation from RoboCommand.

Capabilities Demonstrated:

- Systems must demonstrate the ability to break off from current tasking, travel to the requested location, and then resume overall mission tasking.
- Systems must effectively communicate within the SoS, as well as to operators and judges to provide sufficient situational awareness of the changes in tasking.

Advanced Tier

The RoboCommand system will transmit a static keep out area representing a detected hazard and the impacted domain. Any uncrewed systems operating in the affected domain must acknowledge receipt of this message and avoid the area for the duration of the run or until an *All Clear* message is transmitted by RoboCommand and acknowledgement sent in response by the uncrewed system.

Capabilities Demonstrated:

- Systems must demonstrate the ability to receive and act on messages from RoboCommand while continuing to execute their mission.

Disruptive Tier

The most challenging scenario is an unexpected moving object within the field of operations. During a run, the RoboCommand system will transmit a report of a moving surface object within the operational area and the domain(s) impacted.

All uncrewed systems operating in the impacted domains must avoid approaching within 10m of this object (as visible on the tracking display) based on its current and projected travel. The RoboCommand message will include (at a minimum) GPS location, heading, speed, and system type.

Capabilities Demonstrated:

- The uncrewed system of systems must adapt their mission execution to account for the virtual moving object reported in the area.